

Lecture notes on reflector antennas

Introduction

Reflector antennas are a widely used type of antenna, which are suitable for a multitude of applications all the way from the microwave frequency regime to the visible light. A non-exhaustive list of reflector antenna applications includes astronomical observations (either earth-based or space-based), air traffic control, telecommunication systems (such as satellite TV), imaging for security or health applications and spectroscopic systems. Along with the frequency and specific application, the size of the reflectors is very variable. For earth-based astronomical investigations, telescopes with up to 10m dishes are not uncommon. For high-frequency spectroscopic applications, the reflector size may decrease to a few centimeters or less.

The reason that reflector antennas are so common is that they can provide very high directivity, can be used to tilt a beam and can be used to shape a beam pattern. They are thus flexible tools for the electromagnetic engineer.

In this lecture, we will learn how to evaluate the radiation pattern produced by a reflector, with a reflector feed placed at the focus of the parabolic reflector. We will investigate the trade-offs that can be made in reflector design and in the design of the antenna that feeds the reflector. At the end of this lecture, you will have the knowledge to develop your own (MATLAB) code useful for the analysis of parabolic reflectors.

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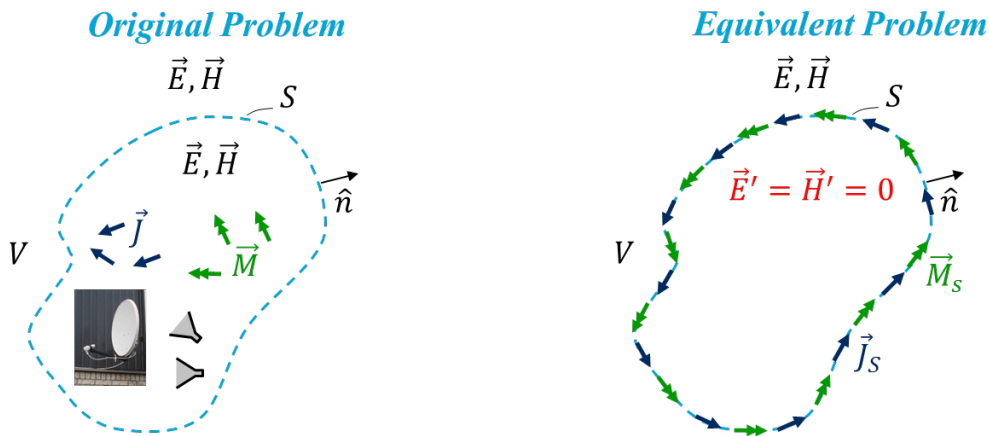
Equivalence theorem (recap)

One of the tools used in the analysis of reflector antennas is the equivalence theorem. With the equivalence theorem we represent electric and magnetic fields inside a closed volume V by equivalent electric and magnetic currents on S , the surface boundary of V .

By imposing a zero field inside the volume V for the equivalent problem, the electric and magnetic current distributions on this surface, $\vec{J}_s, \vec{M}_s$, are given by:

$$\vec{M}_s = \vec{E} \times \hat{n}|_S$$

$$\vec{J}_s = \hat{n} \times \vec{H}|_S$$



Now, the volume V can be filled with an arbitrary material. In Love's formulation of the surface equivalence theorem, the volume V is filled with the same medium of the region external to V (after applying the equivalence theorem). To calculate the radiation from this surface, both the electric and magnetic currents must be taken into account.

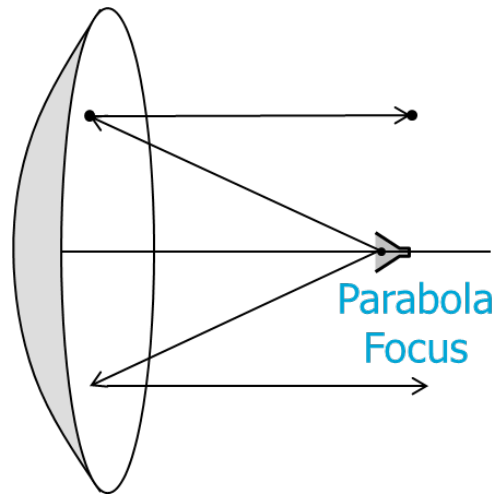
Alternatively, the volume can be filled with a perfect magnetic conductor (PMC). In this case, the magnetic currents are short circuited ($\vec{M}_s = 0$), but for "infinite plane surfaces" using the image theorem the electric currents are doubled: $\vec{J}_s = 2\hat{n} \times \vec{H}|_S$, and radiate in a homogeneous medium. To calculate the radiation outside this volume, only the (doubled) electric currents can be used.

A third approach (not mentioned in these slides) is to fill the volume with a perfect electric conductor (PEC). The PEC short-circuits the electric currents such that only the magnetic currents have to be taken into account when calculating the radiation outside the volume. By applying the image theorem, the magnetic currents are doubled: $\vec{M}_s = 2\vec{E} \times \hat{n}|_S$, and radiate in a homogeneous medium (also in this case an infinite plane surface is assumed).

Parabolic reflector

A parabolic reflector is, as the name suggests, shaped like a parabola. Parabolas are characterized by having a single focus, often called F , which has a peculiar feature: any rays originating from this focal point will be reflected by the parabola in such a way that the rays are parallel to the main axis of the

parabola. Conversely, any rays approaching the parabola parallel to its main axis are focused in the focal point of the parabola. This is (generally) where the antenna is placed.

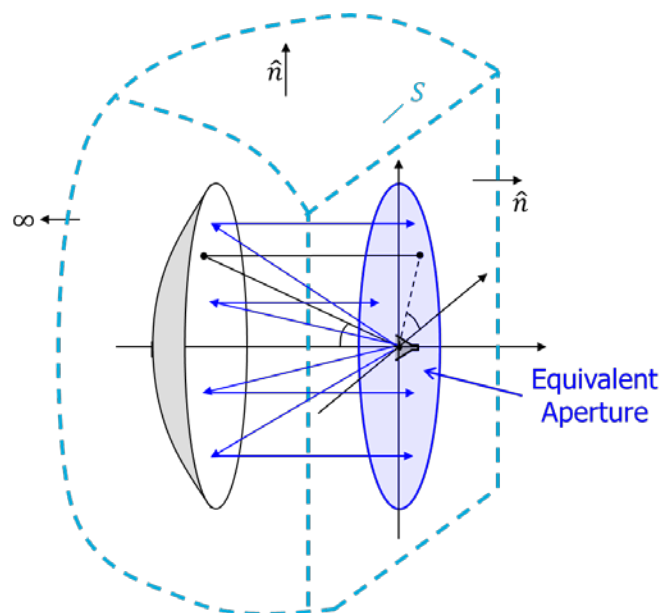


To investigate the radiation from a parabola we make some realistic assumptions about the antenna feeding the parabola and the parabola:

1. The reflector is in the far-field of the antenna, that is, the reflector is electrically far away.
2. The reflector is locally flat with respect to the wavelength, that is, at each point on the reflector we approximate it as a (tilted) flat mirror.
3. The reflector has a uniform reflection coefficient everywhere on the parabola.

Application of the surface equivalence theorem to reflector antennas

The surface equivalence theorem is applied in the following way. First, we define a reference system with its origin at the focal point of the reflector. The direction $\hat{z}$ points away from the parabola. Then, the half-space $z < 0$ is the volume V as intended in the equivalence theorem. The half-space $z > 0$ (facing away from the parabola) is the region in which we would like to evaluate the electric (and magnetic) fields, $\vec{E}, \vec{H}$. The surface S , where the currents significantly contribute to the field scattered by the parabola, is the plane $z = 0$. This is illustrated in the figure below.



To calculate the equivalent surface currents we must know the electric and magnetic field over the surface S . To do this, we make three approximations. The first is the Geometrical Optics (GO) approximation for the fields. This means that we assume that the fields over the aperture are well-approximated by the fields reflected by the parabola or, in formulas, $\vec{E}_a \approx \vec{E}_r$ and $\vec{H}_a \approx \vec{H}_r$. The second approximation is that the equivalent currents are obtained by using these GO fields in the formulas for the equivalent electric and magnetic currents. Third, the fields outside the GO region (the equivalent aperture) are zero. These last two approximations are collectively known as the Physical Optics (PO) approximations.

We assume these three approximations to hold and then it holds that the surface currents can be approximated by:

$$\vec{M}_s \approx \vec{E}_r \times \hat{n}|_S$$

$$\vec{J}_s \approx \hat{n} \times \vec{H}_r|_S$$

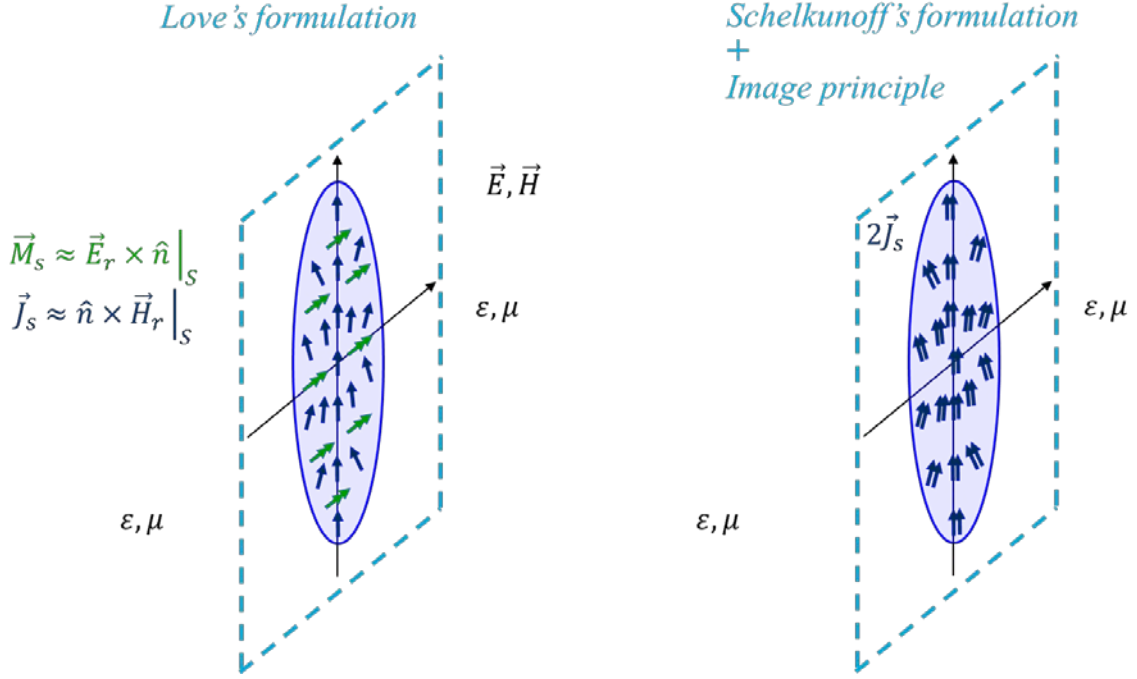
on the equivalent aperture and

$$\vec{M}_s = 0$$

$$\vec{J}_s = 0$$

outside the equivalent aperture.

Finally, if Schelkunoff's formulation (PMC) of the equivalence principle, in combination with the image theorem, is used, the magnetic currents can be set to zero and $\vec{J}_s = 2\hat{n} \times \vec{H}_r|_S$, i.e. the equivalent electric surface currents are doubled by applying the image theorem. This can be seen by filling the half-space $z < 0$ with a PMC. In such case the equivalent problem is valid for $z > 0$. Filling the half-space with PMC is not required, and you may also consider both electric and magnetic currents to be radiating. The figure below shows both possibilities.



Far-field radiation from equivalent currents

In previous lectures about far-field radiation we have seen that there are two approaches to calculating the electric and magnetic fields from current distributions (electric or magnetic): a spectral (asymptotic) approach and a spatial approach. In this lecture we are focusing on the spectral approach.

The spectral method of evaluating the far field is given by the following formula:

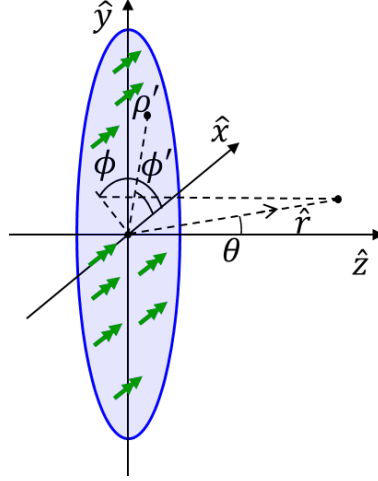
$$\vec{f}^{far}(\vec{r}) \approx jk_{zs} \tilde{\mathcal{G}}^{fc}(k_{xs}, k_{ys}, z, z') \vec{\mathcal{C}}(k_{xs}, k_{ys}) \frac{e^{-jk_r r}}{2\pi r}$$

In which $\vec{f}^{far}(\vec{r})$ represents the field ($\vec{E}$ or $\vec{H}$) that we wish to calculate at some observation point ($\vec{r}$), $\vec{\mathcal{C}}$ is the Fourier transform of the current distribution ($\vec{M}$ or $\vec{J}$) and $\tilde{\mathcal{G}}^{fc}$ is the spectral Green's function of the problem relating the quantity $\vec{f}$ to $\vec{\mathcal{C}}$.

The Fourier transform of the current distribution can be calculated as

$$\vec{\mathcal{C}}(k_x, k_y) = \iint_S \vec{\mathcal{C}}(\vec{r}') e^{jk_x x' + jk_y y'} dS = \iint_S \vec{\mathcal{C}}(\vec{r}') e^{j(k_x x' + k_y y')} dS$$

In general, these far-fields can be calculated by resorting to numerical methods, but some specific current distributions have analytical expressions of their Fourier transform. The example of a uniform circular magnetic current distribution is detailed here. This distribution is characterized by $\vec{M}_s = M_0 \hat{p}_m$ everywhere on a circular domain, centered in the origin, with radius a and $\hat{p}_m$ parallel to the surface, see figure below.



We can now derive the following relationship: $k\hat{r} \cdot \vec{r}' = k\rho' \sin \theta \cos(\phi - \phi')$. This can be used to rewrite the Fourier transform of the magnetic currents:

$$\iint_S \vec{M}(\vec{r}') e^{jk\hat{r} \cdot \vec{r}'} dS = M_0 \hat{p}_m \int_0^a \int_0^{2\pi} e^{jk\rho' \sin \theta \cos(\phi - \phi')} \rho' d\rho' d\phi'$$

The integral in ϕ' above can be closed analytically by using the zeroth-order Bessel identity: $\int_0^{2\pi} e^{jk\rho' \sin \theta \cos(\phi - \phi')} d\phi' = 2\pi J_0(k\rho' \sin \theta)$.

Using this identity leads to:

$$\vec{M}(k_x, k_y) = M_0 \hat{p}_m 2\pi \int_0^a J_0(k\rho' \sin \theta) \rho' d\rho'$$

The integral in ρ' can also be closed analytically by resorting to a property of Bessel functions: $\int t J_0(t) dt = t J_1(t)$ which relates the integral of a zeroth-order Bessel function to the first-order Bessel function. This gives us the following final expression for the Fourier transform of a uniform magnetic current distribution on a circular aperture:

$$\vec{M}(k_x, k_y) = 2M_0 \hat{p}_m \frac{\pi a^2 J_1(ka \sin \theta)}{ka \sin \theta} = M_0 \hat{p}_m 2A \frac{J_1(ka \sin \theta)}{ka \sin \theta}$$

in which $A = \pi a^2$ is the area of the circular surface. This current distribution is called the Airy pattern and it has some interesting properties.

The Airy pattern is rotationally symmetric in ϕ . Its directivity is given by $D = \left(\frac{2\pi a}{\lambda}\right)^2$, which is the maximum directivity that can be achieved by a circular aperture. The zeros of the Airy pattern are given by the zeros of the first-order Bessel function, and the first zero is found at approximately $\frac{69.9\lambda}{2a}$ degrees, its maximum is at broadside ($\theta = 0^\circ$). The relative level of the first sidelobe with respect to the maximum is -17.6 dB. Below, two figures are shown. On the left the Airy pattern is reported in linear scale; whereas, the second shows the Airy pattern in dB scale, normalized to its maximum, with $a = 3\lambda$.

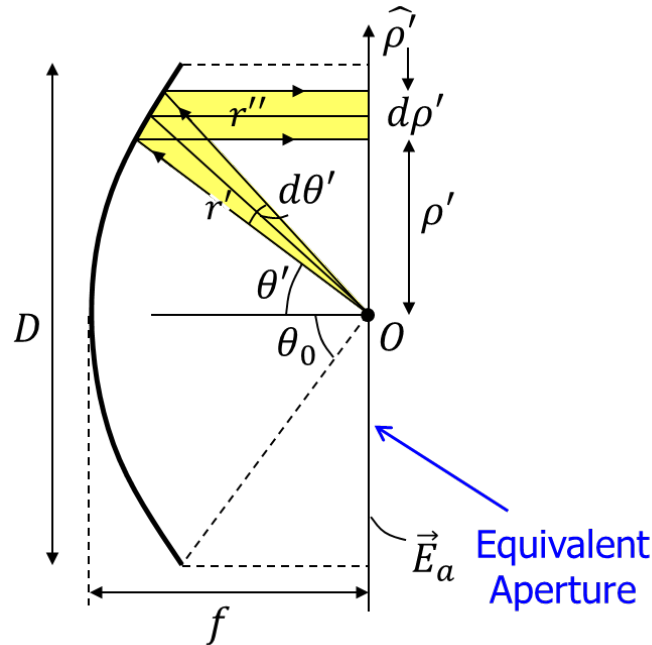
$f \left[1 + \tan^2 \left(\frac{\theta'}{2} \right) \right]$. Furthermore, we have $\rho' = r' \sin \theta' = \frac{2f \sin \theta'}{1 + \cos \theta'} = 2f \tan \left(\frac{\theta'}{2} \right)$. The diameter of the reflector, D , and the subtended half-angle, θ_0 , are related by $D = 4f \tan \left(\frac{\theta_0}{2} \right)$.

Phase of the field on the equivalent aperture

The equivalent aperture field can be represented by three parameters at every point: amplitude, phase, and polarization. The fact that the path length to the equivalent aperture is equal for all rays coming from the focus has two consequences: First, all field contributions on the equivalent aperture will have shifted by the same phase $2kf$. Second, a spherical wave originating from the focus will be transformed to a transverse electromagnetic (TEM) wave propagating along $\hat{z}$. These parabola properties are used to derive the fields over the equivalent aperture.

Amplitude of the field on the equivalent aperture

In the next section we will derive a relation between the amplitude of the feed pattern and the aperture field distribution.



To derive the amplitude distribution, we note that the power density radiated from the feed over a solid angle $d\Omega = \sin \theta' d\theta' d\phi'$ is equal to the power density reflected by the parabola and propagating parallel to the z-axis with the area $dA = \rho' d\rho' d\phi'$. This is shown in the figure above, the power density in the yellow 'tube' is constant. This can be used to derive a relation between the aperture field and the feed pattern distribution:

$$\frac{1}{2\zeta} |\vec{E}_a(\rho', \phi')|^2 dA = U_{feed}(\theta', \phi') d\Omega = \frac{1}{2\zeta} |\vec{f}_{pattern}(\theta', \phi')|^2 d\Omega$$

Where, for the feed, $\vec{f}_{pattern}(\theta', \phi')$ is defined by the relation $\vec{E}(Q) = \vec{f}_{pattern}(\theta', \phi') \frac{e^{-jkr'}}{r'}$ and $\vec{f}_{pattern}(\theta', \phi') = f_{pattern,\theta'}(\theta', \phi') \hat{\theta}' + f_{pattern,\phi'}(\theta', \phi') \hat{\phi}'$.

By substituting the expressions for $d\Omega$ and dA we arrive to:

$$|\vec{E}_a(\rho', \phi')|^2 = 2\zeta U_{feed}(\theta', \phi') \frac{\sin \theta'}{\rho'} \frac{d\theta'}{d\rho'}$$

Next, we use the relation for ρ' that we have seen before to derive $\frac{d\rho'}{d\theta'} = 2f \frac{1}{\cos^2 \frac{\theta'}{2}} \left(\frac{1}{2}\right) = \frac{f}{\cos^2 \frac{\theta'}{2}} = r'$.

Geometrically we can also see that $\frac{\sin \theta'}{\rho'} = \frac{1}{r'}$. These two combined lead to

$$\frac{\sin \theta'}{\rho'} \frac{d\theta'}{d\rho'} = \frac{1}{(r')^2}$$

And

$$|\vec{E}_a(\rho', \phi')| = \sqrt{2\zeta U_{feed}(\theta', \phi')} \frac{1}{r'}$$

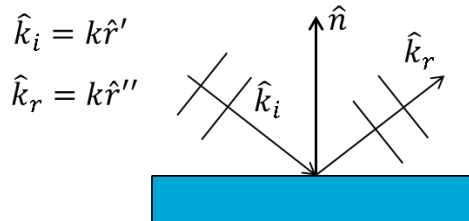
Which is the relation between the amplitude of the feed and the amplitude over the equivalent aperture, as desired. Note that the feed still depends on (θ', ϕ') , as opposed to the aperture distribution that depends on (ρ', ϕ') . We can solve this by using the relation between θ' and ρ' along with $r' = f \left[1 + \left(\frac{\rho'}{2f} \right)^2 \right]$ which we will also use to rewrite $\frac{1}{r'}$:

$$\begin{aligned} |\vec{E}_a(\rho', \phi')| &= \sqrt{2\zeta U_{feed} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right)} \frac{4f}{4f^2 + (\rho')^2} \\ &= \left| \vec{f}_{pattern} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right) \right| \frac{4f}{4f^2 + (\rho')^2} \end{aligned}$$

This means that if the feed radiates a field that has a uniform angular power density, the amplitude of the aperture field distribution is *not* uniform but tapered!

Polarization of the field on the equivalent aperture

The final step in determining the field distribution on the equivalent aperture is to find a relation between the polarization of the feed and the equivalent aperture. To do this, we locally approximate the parabola as a planar PEC interface at each point on the parabola. A plane wave field with propagation direction $\hat{k}_i = k\hat{r}'$ is incident on the interface and a field is reflected in the direction $\hat{k}_r = k\hat{r}''$, as shown in the figure below.



From this problem, the following relations are valid:

1. The components of $\hat{k}_r$ tangent to the planar interface are equal to the corresponding components of $\hat{k}_i$: $\hat{k}_i \times \hat{n} = \hat{k}_r \times \hat{n}$

2. The components of $\hat{k}_r$ orthogonal to the planar interface are opposite to the corresponding components of $\hat{k}_i$: $\hat{k}_r \cdot \hat{n} = -\hat{k}_i \cdot \hat{n}$.
3. The components of $\vec{E}_r$ tangent to the planar interface are equal, but in opposite direction, to the corresponding components of $\vec{E}_i$ and the same is true for the unit vectors of the electric field: $\hat{e}_r \times \hat{n} = -\hat{e}_i \times \hat{n}$
4. The components of $\hat{e}_r$ orthogonal to the planar interface are equal to the corresponding components of $\hat{e}_i$: $\hat{e}_r \cdot \hat{n} = \hat{e}_i \cdot \hat{n}$

1 and 2 determine the propagation direction of the reflected ray: $\hat{r}'' = \hat{r}' - 2(\hat{r}' \cdot \hat{n})\hat{n}$. 3 and 4 determine the polarization of the reflected ray $\hat{e}_r = -\hat{e}_i + 2(\hat{e}_i \cdot \hat{n})\hat{n}$. The normal vector to the parabola is given by $\hat{n} = -\cos\left(\frac{\theta'}{2}\right)\hat{r}' + \sin\left(\frac{\theta'}{2}\right)\hat{\theta}'$ or $\hat{n} = \frac{-\frac{x}{2f}\hat{x} - \frac{y}{2f}\hat{y} + \hat{z}}{\sqrt{\left(\frac{x}{2f}\right)^2 + \left(\frac{y}{2f}\right)^2 + 1}}$.

The following formulas and figure clarify the reference system:

For the **incident field**:

$$\hat{r}' = \frac{Q - F}{|Q - F|}$$

$$\hat{\phi}' = \frac{\hat{n} \times \hat{r}'}{|\hat{n} \times \hat{r}'|}$$

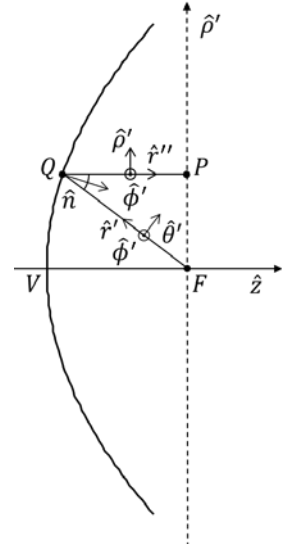
$$\hat{\theta}' = \hat{r}' \times \hat{\phi}'$$

For the **reflected field**:

$$\hat{r}'' = \hat{r}' - 2(\hat{r}' \cdot \hat{n})\hat{n}$$

$$\hat{\phi}' = \frac{\hat{n} \times \hat{r}'}{|\hat{n} \times \hat{r}'|} \equiv \frac{\hat{n} \times \hat{r}''}{|\hat{n} \times \hat{r}''|}$$

$$\hat{\rho}' = \hat{\phi}' \times \hat{r}'' \equiv \hat{\phi}' \times \hat{z}$$



From which one can show that the reflected field consists of two components: a component along $\hat{\phi}'$ that is equal but of opposite sign to the incident field along $\hat{\phi}'$, and a component along $\hat{\rho}'$ that is equal but opposite in sign to the incident field along $\hat{\theta}'$:

$$\hat{e}_r = e_{r\phi'}\hat{\phi}' + e_{r\rho'}\hat{\rho}' = -e_{i\phi'}\hat{\phi}' - e_{i\theta'}\hat{\rho}'$$

Using the more compact dyadic notation, we can define the dyadic $\tilde{R} = -\hat{\rho}'\hat{\theta}' - \hat{\phi}'\hat{\phi}'$ and note that the field reflected in Q is given as:

$$\vec{E}_r(Q) = \tilde{R}(Q) \cdot \vec{E}_i(Q)$$

and the field in P as

$$\vec{E}_r(P) = \vec{R}(Q) \cdot \vec{E}_i(Q) e^{-jkr''} = \vec{E}_r(Q) e^{-jkr''}$$

This can be rewritten in matrix form as:

$$\begin{bmatrix} E_{r,\rho'}(P) \\ E_{r,\phi'}(P) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} E_{i,\theta'}(Q) \\ E_{i,\phi'}(Q) \end{bmatrix} e^{-jkr''}$$

Which shows the desired relationship between the feed polarization and the equivalent aperture polarization. As a final note, remember that the equivalent aperture field is a TEM wave, which means that the magnetic field is related to the electric field by $\vec{H}_r(P) = \frac{1}{\zeta} \hat{r}'' \times \vec{E}_r(P)$.

Calculating the aperture field from the feed field

In the previous three sections, we have seen the relations between, respectively, the phase, amplitude and polarization of the aperture field distribution as a function of the feed field. Here we will combine these three expressions which will result in a final expression linking the feed field to the aperture field.

Phase: We have seen that the relation between the phase on the aperture and the phase of the feed is independent of the location on the aperture; there is always a $2kf$ shift.

Amplitude: The amplitude relation is given by the following equation:

$$|\vec{E}_a(\rho', \phi')| = \left| \vec{f}_{pattern} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right) \right| \frac{4f}{4f^2 + (\rho')^2}$$

Polarization: The aperture field has a component along $\hat{\phi}'$ that is equal but of opposite sign to the incident field along $\hat{\phi}'$, and a component along $\hat{\rho}'$ that is equal but opposite in sign to the incident field along $\hat{\theta}'$.

To conclude, we can write the aperture field components along $\hat{\rho}$ and $\hat{\phi}$ as follows:

$$E_{a,\rho}(\rho', \phi') = -f_{pattern,\theta'} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right) \frac{4f}{4f^2 + (\rho')^2} e^{2jkf}$$

$$E_{a,\phi}(\rho', \phi') = -f_{pattern,\phi'} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right) \frac{4f}{4f^2 + (\rho')^2} e^{2jkf}$$

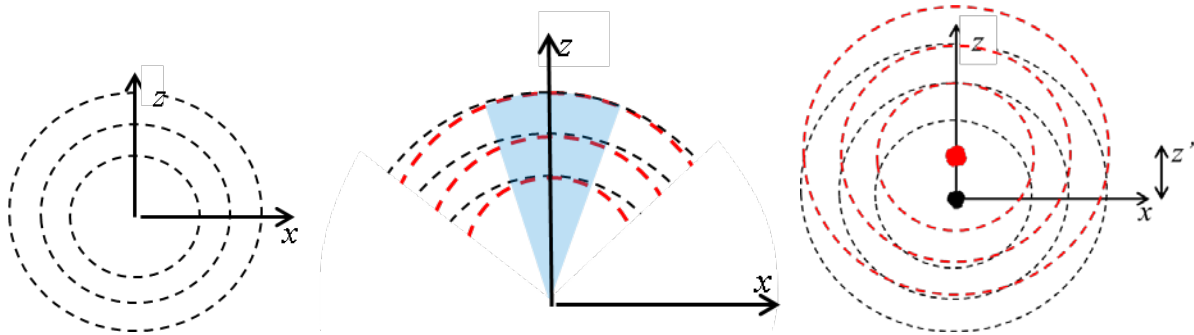
Or combining the vector components:

$$\begin{aligned} \vec{E}_a(\rho', \phi') = & - \left\{ f_{pattern,\theta'} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right) \hat{\rho}' \right. \\ & \left. + f_{pattern,\phi'} \left(2\tan^{-1} \left(\frac{\rho'}{2f} \right), \phi' \right) \hat{\phi}' \right\} \frac{4f}{4f^2 + (\rho')^2} e^{2jkf} \end{aligned}$$

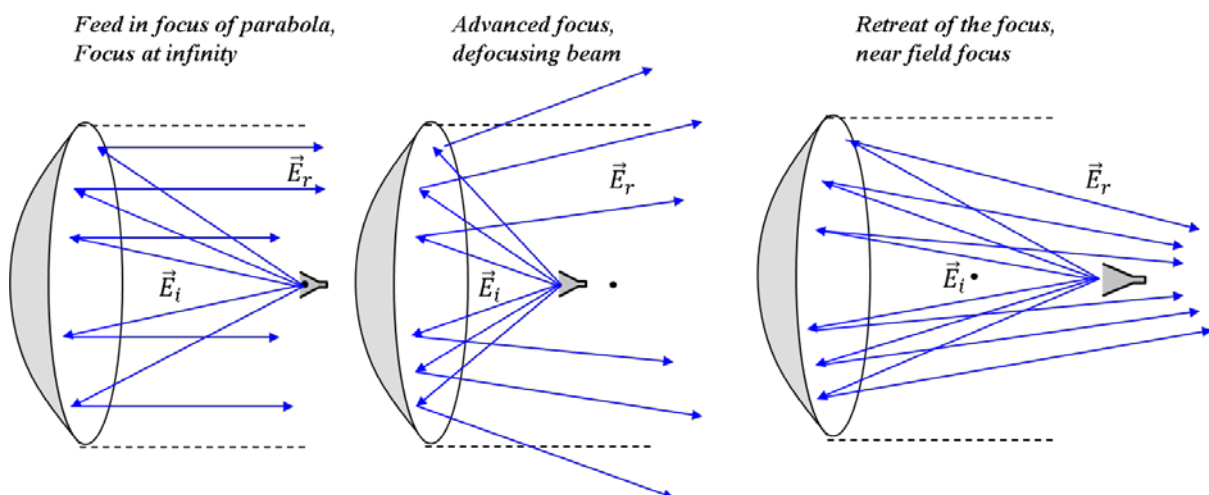
Phase center and practical reflector feeds

In this section, the concept of phase center is discussed. Furthermore, some antennas that are typically used as reflector feeds are discussed and related to their phase center properties.

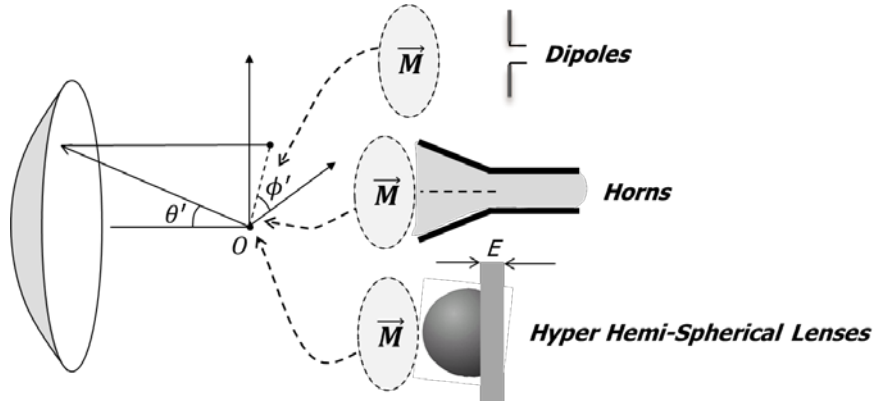
To introduce the phase center, imagine a perfect spherical wave originating from the origin as shown in the left figure below. The point from which the spherical waves originate is called the phase center. In practice, however, antennas don't radiate perfectly spherical waves with a constant phase wavefront. Often the wavefront can be approximated to be spherical and with a constant phase over a limited solid angle, as indicated by the blue region in the middle figure below, which is related to the main beam of the far-field pattern. It can then still make sense to define the phase center as the point from which the spherical wavefront in that (blue region) solid angle seems to originate by minimizing the phase variation in the far-field over that limited solid angle as a function of the presumed phase center location. To minimize this phase variation, a phase term should be included in the feed's radiation pattern to shift the point from which the field is considered from z' to the origin: $e^{\pm jk_z z'}$, where a positive sign is chosen for observation points $z > z'$ and a negative sign is chosen for $z < z'$. This is shown schematically in the left figure below.



The phase center, in practice, does not always coincide with the physical center of the antenna and, thus, care must be taken to place the *phase center* of the antenna in the focus of a reflector, not the physical center. Qualitative consequences of not placing the phase center of the antenna in the focus of the reflector are shown in the figure below.



Now that the phase center has been defined, it is useful to look at antennas typically used as feed for reflectors. Although there are many types of antennas in use, three broad types are described here. They are shown in the figure below.



Dipole antennas are essentially two wires folded open to form a T-like shape. They are resonant antennas (narrow bandwidth due to the resonance-based radiation mechanism) and produce a wide beam. The physical center of the dipole is its phase center and the phase center does not shift with frequency.

Horn antennas result from the flanging of an open-ended waveguide. They have medium directivity and medium operational bandwidth. The phase center of a horn is somewhere in the flanged area of the horn and care must be taken to place it in the reflectors focus. However, the phase center of horns is not constant with frequency which can further limit the operational bandwidth.

Dielectric lens antennas are formed by placing a dielectric lens on top of a radiating structure, such as a slot in an infinite ground plane. There are many variations of lens antennas and each has its own properties: high/low bandwidth and high/low directivity. The phase center of such lens antennas is also dependent on the exact geometry but with careful design the phase center can be kept constant over a relatively large band. However, as with all antennas, the phase center will move when the bandwidths get very large.

Reflector antenna efficiency

To compare the performance of reflectors, one of the main indicators is the aperture efficiency (η_{ap}). The aperture efficiency is the ratio of maximum gain from the reflector to the maximum theoretical directivity from this reflector. We have already seen that the maximum directivity of a circular aperture with radius a is given by $D = \left(\frac{2\pi a}{\lambda}\right)^2$. In general, the maximum directivity of an aperture is given by $D = \frac{4\pi}{\lambda^2} A_{phys}$, whereas the maximum gain is given by $G = \frac{4\pi}{\lambda^2} A_{phys} \eta_{ap}$. In this section we discuss three sources of loss of gain that will lower the aperture efficiency: feed, taper and spillover efficiency.

Feed efficiency (η_f) is the ratio of power radiated by the feed and the power at the input of the feed. There can be many causes of feed inefficiency and these are all grouped together into η_f . The efficiency could, for example, be lower due to mismatching with the connected transmission line, inefficient amplifiers, or radiation in undesired frequencies. For lens antennas specifically, reflection losses at the dielectric-air interface can be considered as feed (in)efficiency. For the current distributions that we are analyzing here, we assume $\eta_f = 1$ unless specified otherwise.

$$\eta_f = \frac{P_{rad}^{feed}}{P_{in}}$$

Taper efficiency (η_t) is a measure of how much the reflector area is effectively used with respect to the physical area projected on the equivalent aperture. It is dependent on the distribution of phase, amplitude, and polarization of the aperture field. It can be calculated from the aperture field as follows:

$$\eta_t = \frac{A_{eff}}{A_{phys}} = \frac{1}{A} \frac{\left| \iint_A E_a^{co}(\rho', \phi') dA \right|^2}{\iint_A |\vec{E}_a(\rho', \phi')|^2 dA}$$

Equivalently, the directivity is the ratio between the achieved directivity to maximum theoretical directivity:

$$\eta_t = \frac{D}{D_{max,theory}} = \frac{D\lambda^2}{4\pi A_{phys}}$$

Alternatively, if the feed gain is given, the *amplitude* efficiency (*Antenna Theory and Design*, Balanis) can be calculated as:

$$\eta_{amp} = \frac{16}{\pi} \left(\frac{f}{D} \right)^2 \frac{\left| \int_0^{2\pi} \int_0^{\theta_0} \sqrt{G_f(\theta', \phi')} \tan\left(\frac{\theta'}{2}\right) d\theta' d\phi' \right|^2}{\int_0^{2\pi} \int_0^{\theta_0} G_f(\theta', \phi') \sin \theta' d\theta' d\phi'}$$

Note that this amplitude efficiency does not include the effects of polarization and phase and therefore does not relate the directivity and the maximum theoretical directivity. If the feed gain is circularly symmetric, the amplitude efficiency reduces to:

$$\eta_{amp} = 32 \left(\frac{f}{D} \right)^2 \frac{\left| \int_0^{\theta_0} \sqrt{G_f(\theta')} \tan\left(\frac{\theta'}{2}\right) d\theta' \right|^2}{\int_0^{\theta_0} G_f(\theta') \sin \theta' d\theta'}$$

Spillover efficiency (η_s) is a measure of the feed power that is intercepted by the reflector: high spillover efficiency means (almost) all the power is intercepted. It can be calculated from feed field distribution:

$$\eta_s = \frac{\int_0^{2\pi} \int_0^{\theta_0} U_{feed}(\theta', \phi') \sin \theta' d\theta' d\phi'}{\int_0^{2\pi} \int_0^{\pi} U_{feed}(\theta', \phi') \sin \theta' d\theta' d\phi'}$$

Where θ_0 is the subtended half-angle of the reflector as seen from the focus.

If the feed gain is given, the spillover efficiency is:

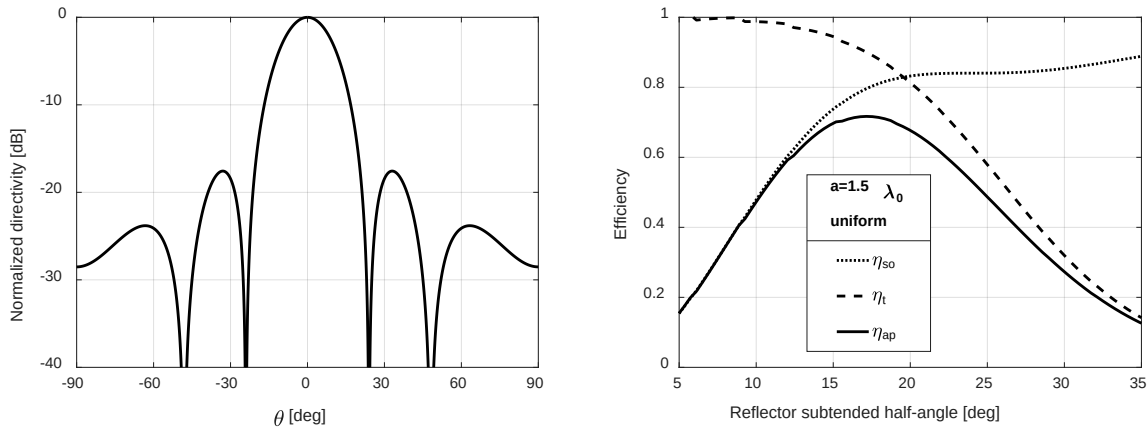
$$\eta_s = \frac{\int_0^{2\pi} \int_0^{\theta_0} G_f(\theta', \phi') \sin \theta' d\theta' d\phi'}{\int_0^{2\pi} \int_0^{\pi} G_f(\theta', \phi') \sin \theta' d\theta' d\phi'}$$

And for circularly symmetric feeds the spillover efficiency is:

$$\eta_s = \frac{\int_0^{\theta_0} G_f(\theta') \sin \theta' d\theta'}{\int_0^{\pi} G_f(\theta') \sin \theta' d\theta'}$$

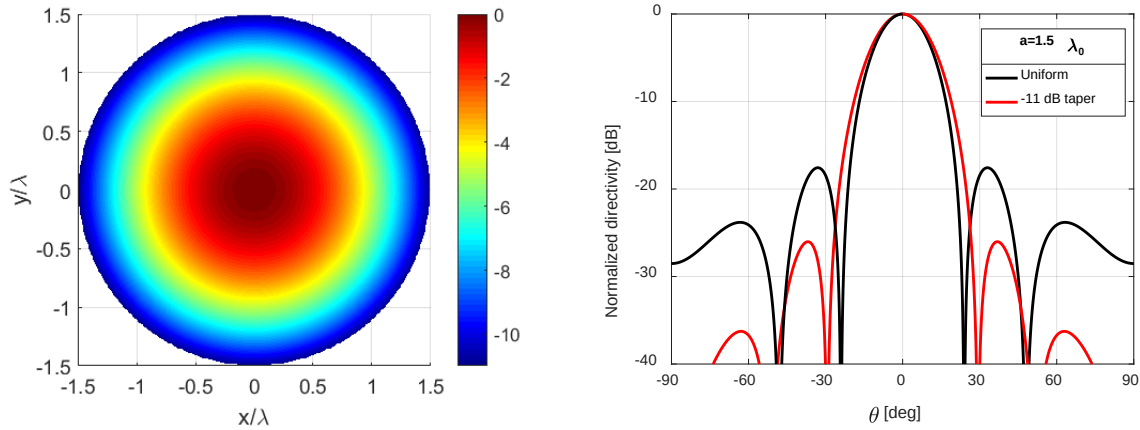
Other efficiencies may be defined as the phase efficiency, polarization efficiency, blockage efficiency (when the reflector is partially blocked) or random error efficiency (surface roughness, reflector imperfections). These are not considered further here but also lead to a decrease in aperture efficiency. The aperture efficiency is the multiplication of all these efficiencies, in the present case $\eta_{ap} = \eta_f \eta_t \eta_s$.

There will always be a trade-off between spillover efficiency and taper efficiency. Consider the case of a uniform circular current distribution with a radius of $a = 1.5\lambda$, with the radiation pattern shown in the figure below, left. The question becomes, how large should we make the reflector in order to maximize the aperture efficiency? To answer this we plot, as a function of the reflector subtended angle, the taper, spillover, and aperture efficiencies (figure below, right). Spillover efficiency increases with subtended angle, while the taper efficiency decreases. Thus, a compromise must be chosen between these two. The aperture efficiency indicates the best reflector dimension to choose has a subtended half-angle of 17.2° which leads to 72% aperture efficiency. This angle corresponds roughly with the -11 dB field level of the feed.

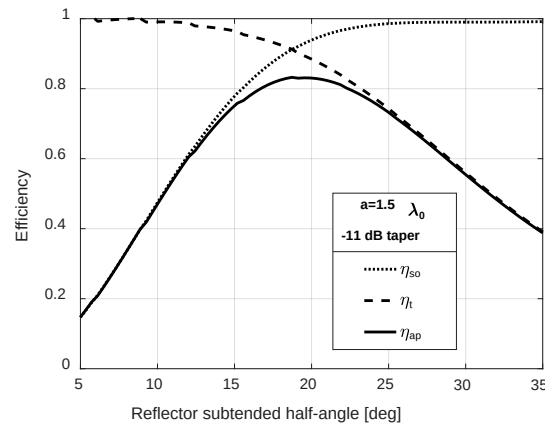


This example shows two important general considerations that arise from uniform circular apertures when used as a feed: First, the maximum aperture efficiency of this reflector-coupled-uniform circular current distribution is about 72%, which means that we are losing the 28% of power no matter what. Second, the angle that corresponds to maximum aperture efficiency is about the -11 dB angle. This angle, of course, will change with the size of the feed.

In the previous example, we have shown the performance (limit) of a uniform circular current distribution. In this next example, we use an exponentially tapered circular current distribution, characterized by $M_{feed} = e^{-\beta(\frac{\rho}{a})^2}$, where we choose $a = 1.5\lambda$ again and $\beta = 1.27$ is a damping constant that we have chosen such that the edge value of the circular distribution is -11 dB relative to the center. The amplitude distribution of the feed is shown in the figure below (left) and the far-field radiation pattern is shown in the figure on the right (in red). We see that the main lobe gets widened and the sidelobe level is reduced.



If we calculate the aperture efficiency of this feed when coupled to a reflector, as a function of the reflector subtended half-angle, the figure below is obtained. The maximum efficiency is now achieved around 19° and this maximum is 82%, much higher than the 72% of the original feed. Again, the maximum is obtained around the -11 dB angle of the feed pattern.



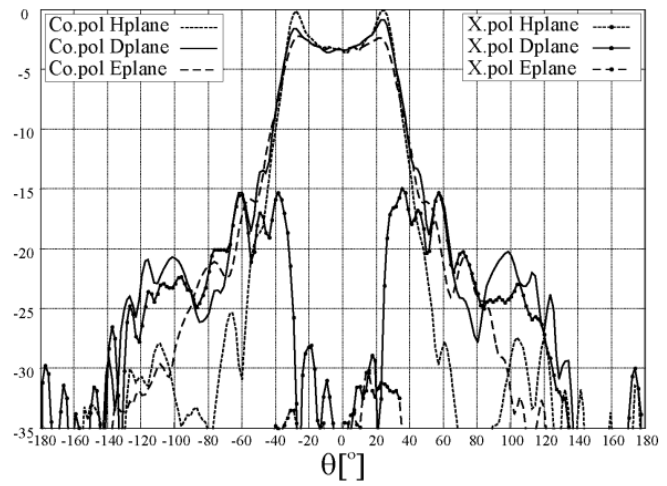
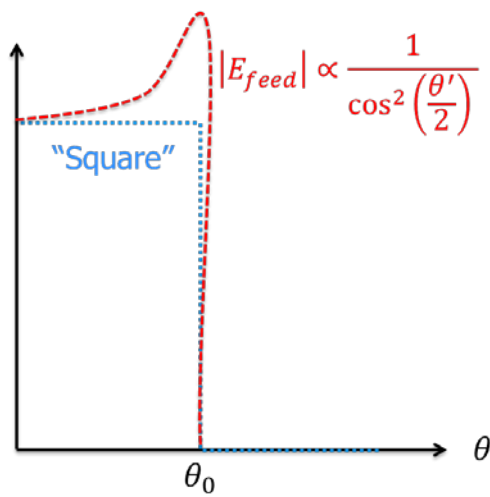
100% aperture efficiency?

Hypothetically, could we achieve 100% aperture efficiency by using a “square” radiation pattern: one that is constant over the angular region of the reflector and zero outside of this region? As we have seen when we analyzed the amplitude distribution of the equivalent aperture of the parabolic

reflector, this aperture field distribution is proportional to $\frac{4f}{4f^2 + (\rho')^2} = \frac{\cos^2\left(\frac{\theta'}{2}\right)}{f}$. To obtain a uniform aperture field distribution, the feed must be

$$|\vec{E}_{feed}| \propto \frac{1}{\cos^2\left(\frac{\theta'}{2}\right)}$$

until the reflector subtended angle. Thus, a “square” radiation pattern will lead to 100% aperture efficiency. The difference is shown schematically in the figure below (left). Such patterns can be obtained by using a so-called “leaky wave antenna”, consisting of layers of dielectrics, to shape the far-field. An example of a measured feed pattern is shown on the right, in which $> 80\%$ aperture efficiency was achieved. Leaky waves will be discussed in detail in the next course, Spectral Domain Techniques.



Further reading:

Books:

Antenna Theory and Design, Balanis (book)